Assignment No. 01

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Implement basic vector and matrix operations in C.

Vector Program Code:-

```
#include<stdio.h>
#include<conio.h>
int main()
{
   int n;
   printf("Enter the Vector size->\n");
   scanf("%d",&n);
   int m[n];
   int m2[n];
   printf("Enter the elements of the Vector 1->\n");
   for(int i=0;i<n;i++)
   {
      scanf("%d",&m[i]);
   }
   printf("Enter the elements of the Vector 2->\n");
   for(int i=0;i<n;i++)
   {
     scanf("%d",&m2[i]);
   }
   printf("Enter your choice->\n1.Sum of Vector\n2.Substraction of Vector\n3.Multiplication
of Vector\n4.Scaling of Vector\n");
   int ch;
   scanf("%d",&ch);
   switch(ch)
```

```
{
    case 1:
        printf("Sum of Vector is->\n");
        for(int i=0;i<n;i++)
        {
            printf("%d",m[i]+m2[i]);
            printf("\n");
        }
        break;
    case 2:
        printf("Substraction of Vector is->\n");
        for(int i=0;i<n;i++)
        {
            printf("%d ",m[i]-m2[i]);
             printf("\n");
        }
        break;
    case 3:
        printf("Multiplication of Vector is->\n");
        for(int i=0;i<n;i++)
        {
            for(int j=0;j<n;j++)
            {
                int sum = 0;
                for(int k=0;k<n;k++)
                {
                    sum+=m[i]*m2[k];
                }
                printf("%d",sum);
                printf("\n");
```

```
            }
            printf("\n");
        }
        break;
    case 4:
        {
            int s;
            printf("Enter the scalar value->\n");
            scanf("%d", &s);
            printf("Scaling of first Vector is->\n");
            for(int i=0;i<n;i++)
            {
                printf("%d",m[i]*s);
                printf("\n");
            }
                printf("\n");
            printf("Scaling of second Vector is->\n");
            for(int i=0;i<n;i++)
            {
                printf("%d",m2[i]*s);
                 printf("\n");
            }
        }
        break;
    default:
        printf("Invalid choice\n");
  }
  return 0;
}
```

Output:

```
PS C:\MFAI> gcc Vector.c -o a;./a.exe
Enter the Vector size->
2
Enter the elements of the Vector 1->
34 5
Enter the elements of the Vector 2->
4 32
Enter your choice->
1.Sum of Vector
2.Substraction of Vector
3.Multiplication of Vector
4.Scaling of Vector
1
Sum of Matrix is->
38
37
```

```
PS C:\MFAI> gcc Vector.c -o a;./a.exe
Enter the Vector size->
2
Enter the elements of the Vector 1->
43 5
Enter the elements of the Vector 2->
4 3
Enter your choice->
1.Sum of Vector
2.Substraction of Vector
3.Multiplication of Vector
4.Scaling of Vector
2
Substraction of Matrix is->
39
2
```

```
Enter your choice->
1.Sum of Vector
2.Substraction of Vector
3.Multiplication of Vector
4.Scaling of Vector
4
Enter the scalar value->
2
Scaling of first Vector is->
66
4
8

Scaling of second Vector is->
46
86
4
```

Matrix Program Code:-

```
#include<stdio.h>
#include<conio.h>
int main()
{
    int r,c;
    printf("Enter the Matrix size->\n");
    scanf("%d%d",&r,&c);
    int m[r][c];
    int m2[r][c];
    printf("Enter the elements of the Matrix 1->\n");
    for(int i=0;i<r;i++)
    {
        for(int j=0;j<c;j++)
        {
            scanf("%d",&m[i][j]);
        }
    }
    printf("Enter the elements of the Matrix 2->\n");
    for(int i=0;i<r;i++)
    {
        for(int j=0;j<c;j++)
        {
            scanf("%d",&m2[i][j]);
        }
    }
    printf("Enter your choice->\n1.Sum of Matrix\n2.Substraction of Matrix\n3.Multiplication
of Matrix\n4.Scaling of Matrix\n");
    int ch;
    scanf("%d",&ch);
    switch(ch)
```

```
{
    case 1:
        printf("Sum of Matrix is->\n");
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
            {
                printf("%d",m[i][j]+m2[i][j]);
                printf("\t");
            }
            printf("\n");
        }
        break;
    case 2:
        printf("Substraction of Matrix is->\n");
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
            {
                printf("%d ",m[i][j]-m2[i][j]);
                printf("\t");
            }
            printf("\n");
        }
        break;
    case 3:
        printf("Multiplication of Matrix is->\n");
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
```

```
        {
            int sum = 0;
            for(int k=0;k<c;k++)
            {
                sum+=m[i][k]*m2[k][j];
            }
            printf("%d",sum);
            printf("\t");
        }
        printf("\n");
    }
    break;
case 4:
    {
        int s;
        printf("Enter the scalar value->\n");
        scanf("%d", &s);
        printf("Scaling of first Matrix is->\n");
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
            {
                printf("%d",m[i][j]*s);
                printf("\t");
            }
            printf("\n");
        }
        printf("Scaling of second Matrix is->\n");
        for(int i=0;i<r;i++)
        {
```

```
                for(int j=0;j<c;j++)
                {
                    printf("%d",m2[i][j]*s);
                    printf("\t");
                }
                printf("\n");
            }
        }
    default:
        printf("Invalid choice\n");
    }
    return 0;
}
```

Output:

```
PS C:\MFAI> gcc Practical_01.c -o a;./a.exe
Enter the Matrix size->
2 2
Enter the elements of the Matrix 1->
12 32
21 4
Enter the elements of the Matrix 2->
32 4
43 4
Enter your choice->
1.Sum of Matrix
2.Substraction of Matrix
3.Multiplication of Matrix
4.Scaling of Matrix
1
Sum of Matrix is->
44      36
64      8
```

```
PS C:\MFAI> gcc Practical_01.c -o a;./a.exe
Enter the Matrix size->
2 2
Enter the elements of the Matrix 1->
32 3
4 5
Enter the elements of the Matrix 2->
2 4
2 5
Enter your choice->
1.Sum of Matrix
2.Substraction of Matrix
3.Multiplication of Matrix
4.Scaling of Matrix
2
Substraction of Matrix is->
30      -1
2       0
```

```
PS C:\MFAI> gcc Practical_01.c -o a;./a.exe
Enter the Matrix size->
2 2
Enter the elements of the Matrix 1->
2 3
1 4
Enter the elements of the Matrix 2->
2 3
4 3
Enter your choice->
1.Sum of Matrix
2.Substraction of Matrix
3.Multiplication of Matrix
4.Scaling of Matrix
3
Multiplication of Matrix is->
16      15
18      15
```

```
PS C:\MFAI> gcc Practical_01.c -o a;./a.exe
Enter the Matrix size->
2 2
Enter the elements of the Matrix 1->
4 3
4 5
Enter the elements of the Matrix 2->
3 4
5 4
Enter your choice->
1.Sum of Matrix
2.Substraction of Matrix
3.Multiplication of Matrix
4.Scaling of Matrix
4
Enter the scalar value->
2
Scaling of first Matrix is->
8       6
8       10
Scaling of second Matrix is->
6       8
10      8
Invalid choice
```